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ICS 2203 WEB APPLICATION DEVELOPMENT 1

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LESSON 10

JAVA SCRIPT

Learning outcomes

Upon completing this topic, you should be able to:

NO OBJECTIVES

10.1. Introduction

What is JavaScript -this is a scripting language that adds programmatic capabilities on the client end. It is an Object Based language unlike its counterpart the normal Java Object oriented language. It solely employs the use of objects, properties of objects and method in programming.



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Similar to other scripting languages JavaScript is light weight. It does not require a compiler because it is an interpreted language.

JavaScript's may be embedded directly into HTML or linked as an external file.

Example of Java Script

```
<HTML>
<body>
<script type="text/JavaScript">
document.write("This is my first JavaScript!");
</script>
</body> </html>
```



10.1.1. Executing JavaScript's

Java scripts usually execute in the context of a webpage on a client's computer. They require a web browser that is java enabled to interpret java scripts embedded in the web page. JavaScript's are renown for altering the appearance of the document, control the display, validate and manipulate form elements, animation of graphics, storing browsing histories and perform general computational tasks.

Having learnt JavaScript however its capabilities will only be limited by your creativity.

The following are some of the capabilities of JavaScript's extracted from the w3c website.

- **JavaScript gives HTML designers a programming tool** - HTML authors are normally not programmers, but



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JavaScript is a scripting language with a very simple syntax! Almost anyone can put small "snippets" of code into their HTML pages.

- **JavaScript can put dynamic text into an HTML page** - A JavaScript statement like this: `document.write("<h1> + name + "</h1>")` can write a variable text into an HTML page.
- **JavaScript can react to events** - A JavaScript can be set to execute when something happens, like when a page has finished loading or when a user clicks on an HTML element.
- **JavaScript can read and write HTML elements** - A JavaScript can read and change the content of an HTML element.



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- **JavaScript can be used to validate data** - A JavaScript can be used to validate form data before it is submitted to a server. This saves the server from extra processing.
- **JavaScript can be used to detect the visitor's browser** - A JavaScript can be used to detect the visitor's browser, and - depending on the browser - load another page specifically designed for that browser.
- **JavaScript can be used to create cookies** - A JavaScript can be used to store and retrieve information on the visitor's computer
- **JavaScript an Object Based Language**

JavaScript is an Object based scripting (OBS) language. It allows you to define your own objects and make your own variable



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types. Similarly it come with a large number of inbuilt objects for usesome of these include the date object, the string object, the window object e.t.c.

In object oriented theories Objects are entities that have properties and are associated with a number of methods. For instance a string object may have a property such as the length.

Consider that example bellow of such an object

```
<script type="text/JavaScript">  
var txt="Welcome To JavaScript's!";  
document.write(txt.length);  
</script>
```

In the following example we are using the length property of the String object to return the number of characters in a string which in this case is 24 counting all the characters in the string:



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Similarly Objects are associated to methods that represent the functions the object will perform. Consider another example of a script that is making use of two methods

In the following example we are using the toUpperCase() method of the String object to display a text in uppercase letters and the write method of the document to display a web output.

```
<script type="text/javascript">  
varstr="Welcome to JavaScript Programming!"; document.write(s  
</script>
```

The output of the code above will be:

WELCOME TO JAVASCRIPT PROGRAMMING!



- **Embedded Java scripts: JavaScript Placement**

To insert a JavaScript into an HTML page, we use the `<script>` tag. Inside the `<script>` tag we use the type attribute to define the scripting language i.e. `<script type="text/JavaScript">` and `</script>` tells where the Java

Script starts and ends:

Everything between `<script>` and `</script>` is processed by the browser as executable JavaScript statements and data. You cannot place HTML or XHTML within this tag; it will be flagged as an error by the browser.

Similar to style sheets we can define JavaScript's in the following ways.

- **Inline scripts:** the script is defined surrounding a portion of content in the body of the web document.



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- **Embedded scripts** : scripts defined in the head portion of web page. Usually they are functions to be executed when needed.
- **External scripts**: scripts defined in a separate file. Use this to centralize script definitions and provide a larger scope across all pages of a web site.

JavaScripts in a page will be executed immediately while the page loads into the browser. This is not always what we want. Sometimes we want to execute a script when a page loads, other times when a user triggers an event.

Scripts in <head>

Scripts to be executed when triggered, go in the head section. You can create some form of a trigger in the body section in the form of controls that when activated will call the scripts in the



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head. It is common practice to have functions in this section. When the page is loaded in memory the script will also load but will not execute until a trigger initiates it

```
<html>
<head>
<script type="text/JavaScript">
function message()
{
  alert("This alert box was called with the onload event resid-
ing in the head section as a function message"); }
</script>
</head>
<body onload="message()">
</body>
```



</html>

Scripts in the Body

We can alternatively place the entire JavaScript code in the body section. The browser in this case will automatically execute all the content of the script tag without requiring a further trigger.

The script in this particular case will be responsible for the browser content.

```
<html>
```

```
<head>
```

```
</head>
```

```
<body>
```

```
<script type="text/JavaScript">
```

```
document.write("This message is written by JavaScript");
```



</script>

</body>

</html>

10.1.2. JavaScript Objects

Because most of the JavaScript code that you write involves objects, properties, methods, and event handlers, understanding what these object-oriented terms mean is essential for folks planning to write their own scripts.

You can think of it this way:

- Objects are always nouns.
- Properties are adjectives.
- Methods are verbs.
- Event handlers are verbs with on tacked to their fronts



- **The Object Hierarchy**

The object hierarchy: this hierarchy applies to the organization of objects in a document.

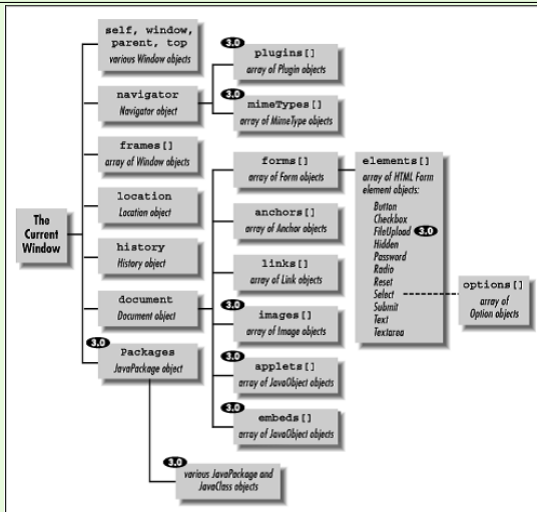
Window Object: this appears at the top of the hierarchy representing the content area of the browser. In multiple frame environments each frame is a window

The Document Object: each HTML document that gets loaded into a window becomes a document object. The Document Object contains most of other kinds of objects in the model

The Form Object: a form is a distinct grouping of content inside HTML ie everything inside the `<form>...</form>` element. A document may have two or more form elements.



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Dates and times are not fundamental types. But JavaScript does provide a type (or class) of object that represents dates and



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times, and can be used to manipulate this type of data. A Date object in JavaScript is created with the new operator and the Date() constructor:

```
now = new Date(); // create an object representing the current
```

date and time

```
xmas = new Date(96, 11, 25); // Create a Date object
```

representing Christmas

representing Christmas

Methods of the Date object allow you to get and set the various date and time values, and to convert the Date to a string, using either local time or GMT time. For example:

```
xmas.setYear(xmas.getYear() + 1); // Change the date to next Christmas
```



```
document.write("Today is: " + now.toLocaleString());
```

In addition, the Date object also defines functions (not methods; they are not invoked through a Date object) to

convert a date specified in string or numeric form to an internal millisecond representation that is useful for some kinds of date arithmetic.

10.1.3. Java Script Coding

- JavaScript code (or just JavaScript) is a sequence of JavaScript statements.
- Each statement is executed by the browser in the sequence they are written.
- This example will write a heading and two paragraphs to a web page:



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```
<script type="text/javascript">  
    document.write("<h1>This is a heading</h1>");  
    document.write("<p>This is a paragraph.</p>");  
    document.write("<p>This is another paragraph.</p>");  
</script>
```

JavaScript statements can be grouped together in blocks.

Blocks start with a left curly bracket {, and ends with a right curly bracket }.

The purpose of a block is to make the sequence of statements execute together.

This example will write a heading and two paragraphs to a web page:

```
<script type="text/javascript">  
{
```



```
document.write("<h1>This is a heading</h1>");  
document.write("<p>This is a paragraph.</p>");  
document.write("<p>This is another paragraph.</p>");  
}  
</script>
```

10.1.4. Variables

- JavaScript variables are used to hold values or expressions.
- A variable can have a short name, like x, or a more descriptive name, like carname.

Rules for JavaScript variable names:

- Variable names are case sensitive (y and Y are two different variables)



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- Variable names must begin with a letter or the underscore character
- Note: Because JavaScript is case-sensitive, variable names are case-sensitive.
- Creating variables in JavaScript is most often referred to as "declaring" variables.
- You can declare JavaScript variables with the var statement:

var x;

varcarname;

After the declaration shown above, the variables are empty (they have no values yet).

However, you can also assign values to the variables when you declare them:



```
var x=5;  
varcarname="Volvo";
```

10.1.5. Java Script Functions

Similar to other programming a function is a small program that performs given tasks.

To keep the browser from executing a script when the page loads, you can put your script into a function.

A function contains code that will be executed by an event or by a call to the function.

You may call a function from anywhere within a page (or even from other pages if the function is embedded in an external .js file).

Functions can be defined both in the <head> and in the



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<body> section of a document. However, to assure that a function is read/loaded by the browser before it is called, it could be wise to put functions in the <head> section.

Syntax

```
functionfunctionname(var1,var2,...,varX)
```

```
{
```

```
some code
```

```
}T
```

he parameters var1, var2, etc. are variables or values passed into the function. The { and the } defines the start and end of the function

```
<html>
```

```
<head>
```

```
<script type="text/javascript">
```



```
function displaymessage()  
{  
  alert("Hello World!");  
}  
</script>  
</head>  
<body>  
  <form>  
    <input type="button" value="Click me!"  
    onclick="displaymessage()" />  
  </form>  
</body>  
</html>
```

Example 2



```
<html>
<head>
<script type="text/javascript">
function product(a,b)
{
return a*b;
}
</script>
</head>
<body>
<script type="text/javascript">
document.write(product(4,3));
</script>
</body>
```



</html>

10.1.6. Javascript Pop up Boxes

The hallmark of JavaScript lies in the ability of javascript pop ups

- **Alert Box**

- An alert box is often used if you want to make sure information comes through to the user.
- When an alert box pops up, the user will have to click "OK" to proceed.

Syntax

alert("sometext"); i.e.

<html>



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```
<head>
<script type="text/javascript">
function show_alert()
{
alert("I am an alert box!");
}
</script>
</head>
<body>
<input type="button" onclick="show_alert()" value="Show
alert box" />
</body>
</html>
```

Confirm box



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- A confirm box is often used if you want the user to verify or accept something.
- When a confirm box pops up, the user will have to click either "OK" or "Cancel" to proceed.
- If the user clicks "OK", the box returns true. If the user clicks "Cancel", the box returns false.

Syntax

confirm("sometext"); i.e.

```
<html>
```

```
<head>
```

```
<script type="text/javascript">
```

```
function show_confirm()
```

```
{
```

```
var r=confirm("Press a button");
```



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```
if (r==true)
```

```
{
```

```
document.write("You pressed OK!");
```

```
}
```

```
else { document.write("You pressed Cancel!");
```

```
}
```

```
}
```

```
</script>
```

```
</head>
```

```
<body>
```

```
<input type="button" onclick="show_confirm()" value="Show
```

confirm

```
box" />
```

```
</body>
```



</html>

- **Prompt box**

A prompt box is often used if you want the user to input a value before entering a page.

When a prompt box pops up, the user will have to click either "OK" or "Cancel" to proceed after entering an input value.

If the user clicks "OK" the box returns the input value. If the user clicks "Cancel" the box returns null.

Syntax

```
prompt("sometext","defaultvalue");
```



10.1.7. JavaScript event Handlers

- One of the most important features provided by JavaScript is the ability to detect and react to events that occur while a document is loading, rendering, and being browsed by the user. The JavaScript code that handles these events may be placed within the `<script>` tag, but more commonly, it is associated with a specific tag via one or more special tag attributes.
- For example, you might want to invoke a JavaScript function when the user passes the mouse over a hyperlink in a document. The JavaScript-aware browsers support a special "mouse over" event-handler attribute for the `<a>` tag called `onMouseOver` to do just that:



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- `<a href="doc.html" onMouseOver="status='Click me!'; return true">`
- `onAbort*`
- `<img>`
- `onBlur`
- `<a><area><body><button><frameset>`
 - `<input><label><select><textarea>`
- `onChange`
- `<input><select><textarea>onClickListener`
 - `DbClickMosttagsonError*`
- `<img>`
- `onFocus`

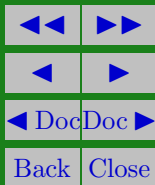


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- `<a><area><body><button><frameset>`
 - `<input><label><select><textarea>`
- `onKeyDownMost- tagsonKeyPressMost tags`
- `onKeyUp-Most tagsonLoad`
- `<body><frameset><img>*`
- `onMouseDown- Most tags`
- `onMouseMove -Most tags`
- `onMouseOut-Most tags`
- `onMouseOver-Most tags`
- `onMouseUp -Most tags`
- `onReset`
- `<form>onSelect`



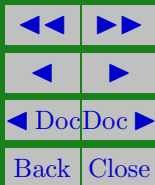
- `<input>``<textarea>`
- `onSubmit`
- `<form>``onUnload`
- `<body>``<frameset>`
- **The mouse-related events**
 - The `onClick`, `onDblClick`, `onMouseDown` and `onMouseUp` attributes refer to the mouse button.
 - If you need to detect both halves of a mouse click as separate events, use `onMouseDown` and `onMouseUp`. When the user presses the mouse button, the `onMouseDown` event occurs. The `onMouseUp` event happens when the user releases the mouse button.





- The onMouseMove ,onMouseOut, and onMouseOver events happen when the user drags the mouse pointer. The onMouseOver event occurs when the mouse first enters the display region occupied by the associated HTML element. After entry, onMouseMove events are generated as the mouse moves about within the element. Finally, when the mouse exits the element, onMouseOut occurs.
- For some elements, the onFocus event corresponds to onMouseOver, and onBlur corresponds to onMouseOut.
- **The keyboard events**

Only three events currently are supported by the HTML 4 and XHTML standards relating to user keyboard actions: onKeyDown ,onKeyUp and onKeyPress,. The onKeyDown event oc-





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curs when the user depresses a key on the keyboard; onKeyUp happens when the key is released. The onKeyPress attribute is triggered when a key is pressed and released. Usually, you'll have handlers for either the up and down events, or the composite keypress event, but not for both.

- **Document events**

Most of the document-related event handlers relate to the actions and states of form controls. For instance, onReset and onSubmit happen when the user activates the respective reset or submit button. Similarly, onSelect and onChange occur as users interact with certain form elements. There also are some document-related event handlers that occur when various document elements get handled by the browser. For instance, the





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onLoad event may happen when a frameset is complete, or when the body of an HTML or XHTML document gets loaded and displayed by the browser. Similarly, onUnload occurs when a document is removed from a frame or window.

Example . Write a JavaScript statement to accomplish each of the following tasks

Declare variables sum and x.

Assign 1 to variable x.

Assign 0 to variable sum

Add variable x to variable sum and assign the result to variable sum

Print “The sum is” followed by the value of variable sum

Solution: i. Declare variables sum and x. `var x; var sum;` ii. Assign 1 to variable x. `x=1 ;` or `var x = 1;` iii. Assign 0 to



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variable sum. sum =0 ; or var sum =0; iv. Add variable x to variable sum and assign the result to variable sum. sum = xAssignments + sum; v. Print “The sum is” followed by the value of variable sum. document.write(“The sum is”+sum);

Learning Activities

Revision Questions

EXERCISE 1.  Giving examples, explain the following terms

Objects

Properties

Methods

EXERCISE 2.  Describe the role of JavaScript in HTML



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EXERCISE 3. ✎ Write a simple script that demonstrates the role stated above

EXERCISE 4. ✎ Highlight any four events that may be captured by JavaScript

References and Additional Reading Materials

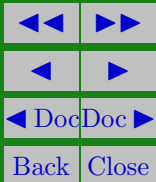
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Solutions to Exercises

Exercise 1. i. Objects an entity that has a state behavior and identity ie a window object in javascript ii. Properties these re attributes of an object ie the length of a string object iii. Methods there are the behaviors associated with an object i.e. .write () method of a document object in javascript

Exercise 1

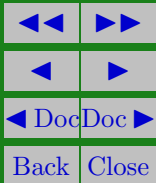


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Exercise 2. Put answer

Exercise 2





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Exercise 3. Put answer

Exercise 3

◀◀	▶▶
◀	▶
◀ Doc	Doc ▶
Back	Close



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Exercise 4. Put answer

Exercise 4

